DNN Programming Assignment 2

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Problem 1: data, verification and HAR classification

UCI HAR uses the prescribed nine inertial channels, 128 steps, six activities, 7352 source-training windows and 2947 official test windows. The 21 source-training subjects and nine test subjects are disjoint. For each seed 0/1/2, four training subjects are selected by the prescribed NumPy choice for validation. Channel normalization uses only the remaining training subjects. Figure 1 shows the splits; Figure 2 shows the validation-selected confusion matrix. All classifiers use last-state classification, $h = 64$, one recurrent layer, Adam 10^{-3}, batch 128, ten epochs and norm-1 clipping. Validation macro-F 1 selects checkpoints; reported spreads are sample standard deviations.

The scratch cells match library outputs within 10^{-5}; assert cells also verify backward gradients and nonzero initial states. For $g = 1, 4, 3$ gates, the single-bias count is $gh(d + h + 1)$ and PyTorch's two-bias count is $gh(d + h + 2)$. At $d = 9, h = 64$, RNN/LSTM/GRU counts are 4736/18944/14208 versus 4800/19200/14400. Classifier heads add $6h + 6$.

Table 1. HAR: three seeds; GTX 1650 training time

Model	Acc. (%)	Macro-F 1	Params	s/epoch
GRU	88.0783 ± 0.6369	0.8807 ± 0.0063	14790	1.419
LSTM	87.5014 ± 0.4243	0.8750 ± 0.0046	19590	1.432
RNN	72.4579 ± 1.0483	0.7159 ± 0.0085	5190	0.581

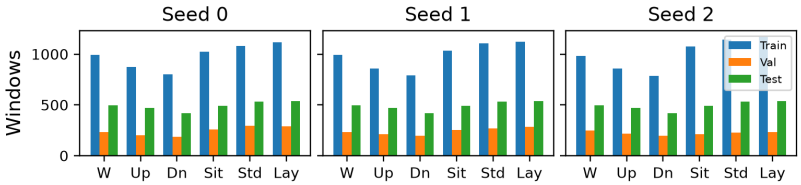


Figure 1. Class balance by subject split and official test set.

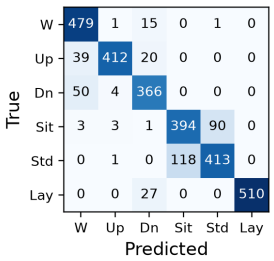


Figure 2. Validation-selected GRU, seed 2, epoch 9. Table 1 compares all architectures.

Table 2 keeps the same HAR budget and subject splits while varying hidden width.

Table 2. HAR hidden-size sweep

h	LSTM acc. (%)	Params	GRU acc. (%)	Params
16	76.2697 ± 6.6941	1830	78.7694 ± 2.0219	1398
32	83.8254 ± 1.8710	5702	87.3317 ± 1.1564	4326
64	87.5014 ± 0.4243	19590	88.0783 ± 0.6369	14790
128	87.7276 ± 0.6798	71942	88.9492 ± 0.1930	54150

Problem 1: adding, width and individual ablation

The prescribed adding generator uses two markers, one per half; assertions check marker counts and exact target sums. Each length uses 10000/1000/1000 examples with data seeds 0/1/2. All models use $h = 64$, Adam 10^{-3}, batch 128, 50 epochs and clipping 1, with three model seeds. Validation MSE selects weights. The theoretical constant-1 baseline is 1/6; Table 3 uses the measured baseline. Solved means test MSE below 0.05; Figure 3 contains length and learning curves.

Table 3. Adding MSE; solved R/L/G counts out of three

T	RNN	LSTM	GRU	Constant 1	Solved
50	0.116151 ± 0.082032	0.002012 ± 0.000542	0.000238 ± 0.000058	0.1754	1/3/3
100	0.169745 ± 0.000937	0.025186 ± 0.031006	0.000581 ± 0.000057	0.1697	0/2/3
200	0.168346 ± 0.000209	0.168243 ± 0.000256	0.000493 ± 0.000190	0.1679	0/0/3
400	0.153994 ± 0.000453	0.153601 ± 0.000043	0.001764 ± 0.000396	0.1527	0/0/3

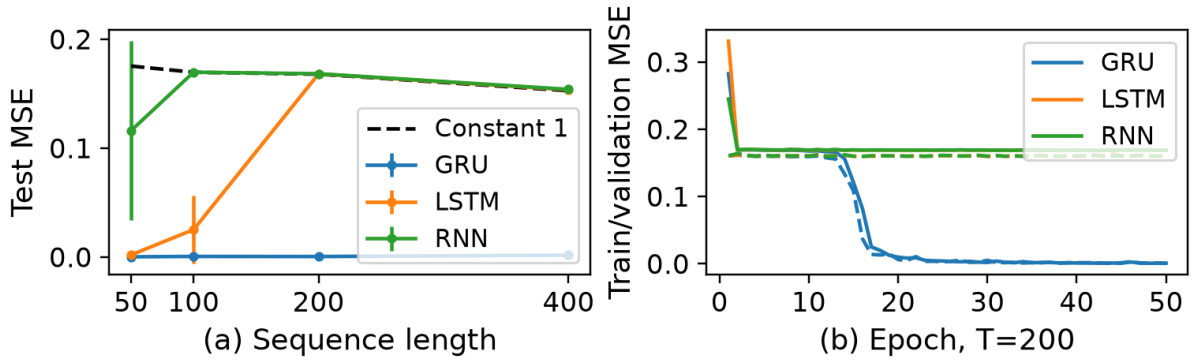


Figure 3. Test MSE against length; T=200 curves: solid training, dashed validation.

ID ends in 3, so the required individual experiment compares initial LSTM forget bias 0/1 at T=200 (Table 4); other initial weights and shuffle seeds are paired. Figure 4 shows this and the hidden-size study (Table 2).

Table 4. Forget-bias ablation, three seeds

Bias	MSE	Solved
0	0.166789 ± 0.002848	0/3
1	0.167462 ± 0.003357	0/3

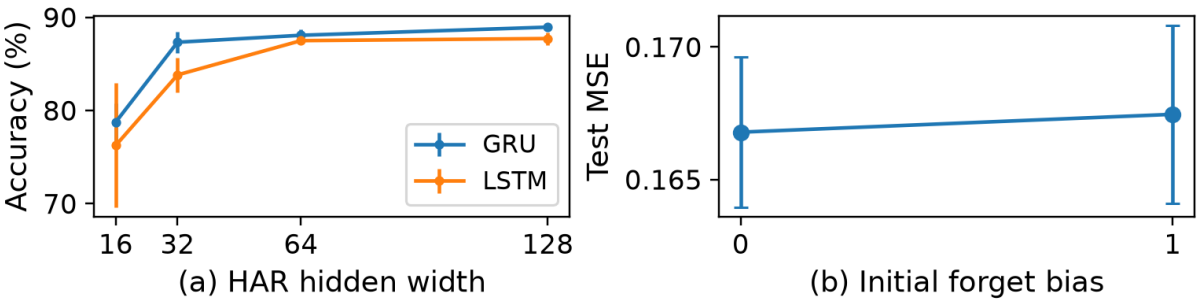


Figure 4. Hidden-size trade-offs and initial forget-bias comparison.

Problem 1: gradient evidence and one-page analysis

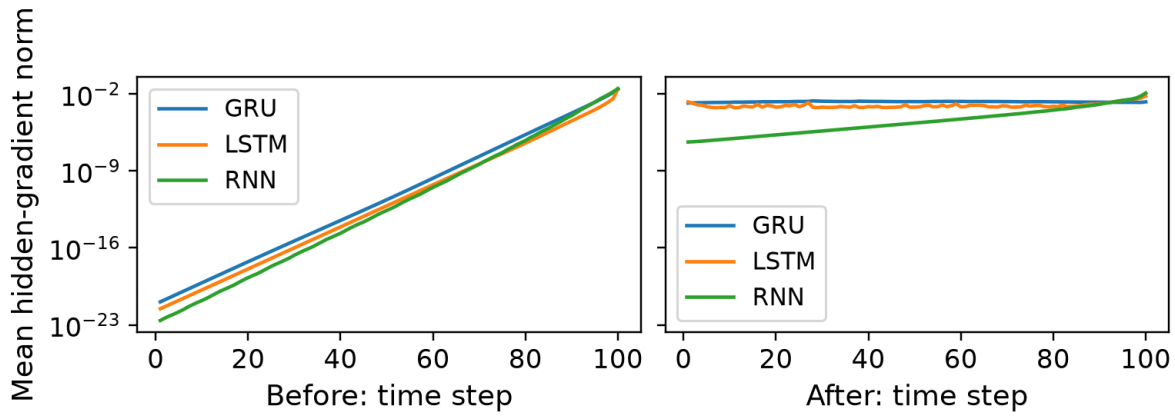


Figure 5. Actual hidden-state gradient norms at T=100 before/after training; logarithmic scale, three-seed means.

A fixed independent seed 3 probe contains 32 examples. Float 64 custom unrolls retain each hidden state and verify agreement with library predictions and parameter gradients. LSTM retains a separate cell-state graph. Small early hidden gradients indicate weak direct credit assignment to early states, but do not alone measure all memory carried through an LSTM cell state. Figure 5 should therefore be read alongside the predictive evidence in Table 3, not as proof that gates solve every task. RNN: mean post-training first/last hidden-gradient norms are $4.33 \times 10^{-7}/1.23 \times 10^{-2}$. LSTM: mean post-training first/last hidden-gradient norms are $1.92 \times 10^{-3}/6.61 \times 10^{-3}$. GRU: mean post-training first/last hidden-gradient norms are $1.54 \times 10^{-3}/1.95 \times 10^{-3}$.

Table 3 records 18/36 solved runs under a common 50-epoch budget. Validation-only T=50 pilots motivated this frozen follow-up budget after the earlier five-epoch grid solved none. Earlier test results were already observed; this is not a new untouched holdout. Validation MSE selects each checkpoint. GRU solves all lengths; LSTM becomes unstable at T=100 and stalls at T=200/400; RNN solves only one short run. This is finite-budget evidence, not architectural impossibility.

Table 4: neither initial forget bias solves T=200. Paired bias 1-minus-bias 0 MSE differences are 0.00135, 0.00054, 0.00013 (mean 0.00067, sample std 0.00062). Negative favors bias 1; no benefit is observed here. Baseline-level losses limit conclusions about forgetting or a universal ranking.

Table 1: GRU uses 24.5 percent fewer parameters than LSTM, with 0.58 percentage-point accuracy difference. Mean training seconds/epoch are 0.581/1.432/1.419 for RNN/LSTM/GRU. A smaller or faster model is useful only if its measured accuracy is acceptable. For a wearable prototype I would start with GRU, selected by the highest mean validation F 1 (0.9075), then consider smaller widths in Table 2 if memory is constrained. Table 1 supplies its accuracy, parameters and training cost; GPU training times do not measure wearable inference latency or energy. Deployment needs on-device validation. Local NVIDIA GeForce GTX 1650 (4 GB): independently fresh P 1/final P 2 workflows took 96.27 minutes before this presentation revision, including tests, training, evaluation, plots and CPU diagnostics; setup, downloads and an earlier P 2 optimization attempt are excluded. Current cached verification is separate. T4 timing was not measured. The KV-cache bonus is included.

AI-use disclosure. OpenAI Codex assisted with scratch implementations, training, testing, execution, plots, analysis and report preparation. Claude supplied an external review. The current assignment permits AI assistance for scratch components.

Problem 2: scratch Transformer and HAR encoder

Attention is $A = \mathrm{softmax}(QK^T/\sqrt{d_k} + M)$, $O = AV$. Assertions verify row sums, shapes, zero future weights, gradients and agreement with PyTorch SDPA within 10^{-5}. Framework attention/encoder layers are used only in tests. Fully masked rows are rejected. A nonuniform hand example substitutes for the absent instructor fixture: $Q = [(1,0);(0,1);(1,1)]$, $K = [(1,0);(0,1);(1,-1)]$, $V = [(1,0);(0,2);(3,4)]$. For $a = 1/\sqrt{2}$, score rows are $(a,0,a),(0,a,-a),(a,a,0)$. Exponentiate and divide each row by its sum; outputs are $(1.6044, 2.0000),(0.7041, 1.7121),(0.9944, 1.5933)$, agreeing with code.

Q/K/V/O projections reshape into heads. Pre-LayerNorm blocks use residuals, ReLU FFN width$4d$ and dropout. Sinusoidal positions use $PE_{p,2i} = \sin(p/10000^{2i/d})$ and cosine for odd components. Matrix weights per block are $4d^2 + 2dd_{ff}$:196608 at $d = 128, d_{ff} = 512$. Biases and LayerNorm add $13d = 1664$, total 198272, checked by summing parameters. For the HAR encoder, $d = 64$:16384+32768=49152 weights, plus 832 gives 49984.

TaskA uses 8-step patches, a CLS token, $d = 64$, two blocks/four heads, dropout 0.1, batch 128 and ten epochs under the same subject splits. Table 5 and Figure 6 compare no, sinusoidal and learned positions. Recurrent baselines are in Table 1; patching, dropout and parameter counts confound architecture-only conclusions.

Table 5. HAR encoder, three seeds

Position	Val F 1	Test acc. (%)	Test F 1	Params	s/epoch
learned	0.9190	91.5847 ± 2.0396	0.9159 ± 0.0201	106310	0.308
none	0.9274	89.5374 ± 0.7666	0.8945 ± 0.0076	105222	0.294
sinusoidal	0.9314	90.3970 ± 1.1095	0.9044 ± 0.0118	105222	0.284

The highest mean validation F 1 (0.9314) selects sinusoidal, rather than choosing by test accuracy. Against the validation-selected GRU in Table 1, it achieves 90.40 versus 88.08 percent test accuracy (+2.32 percentage points), using 105222 versus 14790 parameters (7.11 times). Mean local training time is 0.284 versus 1.419 s/epoch. Patching shortens attention to 17 tokens versus 128 recurrent steps; size and dropout also differ, so these timings do not isolate architecture or measure energy.

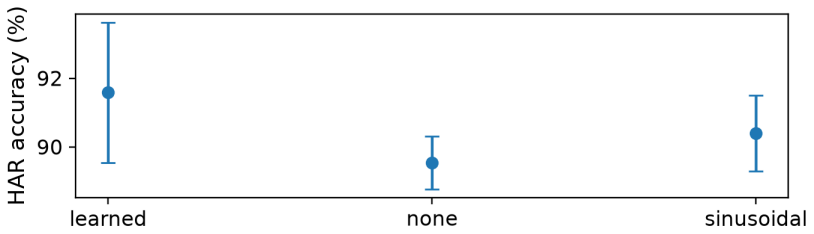


Figure 6. Position ablation; error bars are sample standard deviations.

Removing positions makes attention invariant to permutations of whole patches, verified by reversing patch order. It does not remove order within each flattened 8-step patch. Thus a strong no-position result would not prove temporal order irrelevant. Sizes/seeds are disclosed assumptions where the variant table is missing.

Problem 2: causal language model and ablations

Tiny Shakespeare uses its full 65-character vocabulary; the first 90 percent trains and last 10 percent validates, without shuffled text. The standard GPU model uses $d = 128$, four layers, context 128, four heads, batch 64, dropout 0.1, Adam 10^{-3} and five epochs. Heads 1/4/8 and mask removal have matched budgets. Four additional five-epoch continuations reset Adam consistently from each selected baseline. This deliberate restart protocol does not isolate duration from restart effects.

Table 6. LM validation; BPC = nats / ln 2

Heads	Mask	Base nats	Continued nats	BPC
4	causal	1.9093	1.7444	2.5167
1	causal	1.8971	1.7538	2.5302
8	causal	1.9226	1.7547	2.5315
4	unmasked	0.0285	0.0210	0.0303

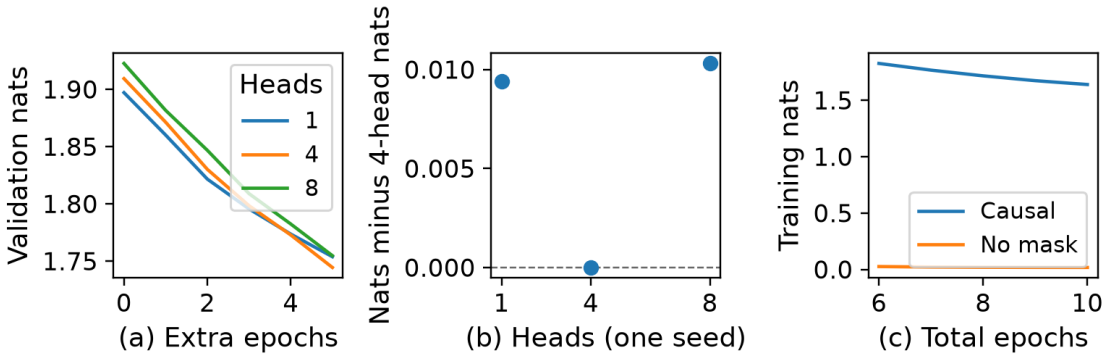


Figure 7. Validation learning; head-loss difference from four heads (one seed); mask-removal training loss.

Uniform prediction costs $\ln 65 = 4.17439$ nats. Table 6 and Figure 7 provide learning evidence; one seed does not establish a reliable head-count ranking. Unmasked training exposes the next-character target through future input tokens: its validation loss is contaminated and is not autoregressive quality. Causal-prefix invariance and an unmasked counterexample are tested on the trained weights. The primary LM is the continued four-head causal checkpoint. All samples contain 500 new characters after `ROMEO:\n`, top-k 20, sampling seed 700.

Table 7. Sample diversity: continued main model; unmasked baseline

Temperature/model	Distinct characters	Empirical entropy
0.2	26	3.673
0.8	41	4.379
1.5	47	4.687
unmasked,0.8	1	-0.000

The baseline unmasked draw at 0.8 has 100.0 percent newlines and 100.0 percent identical adjacent characters (Table 7). Compare this generation evidence with its contaminated validation loss in Table 6. Table 7 describes individual draws, not model predictive entropy. Temperature 0.2 concentrates probability on likely characters; 0.8 permits more variation; 1.5 flattens the distribution. Higher character diversity is not evidence of coherent language. At 0.2, repeated ‘shall’ fragments dominate; 0.8 contains speaker-like ‘POMPENEy’ but malformed ‘lannio’; 1.5 varies punctuation and case while misspelling ‘wildrak’. The full labelled samples are in generated_samples.txt.

Problem 2: attention interpretation and context scaling

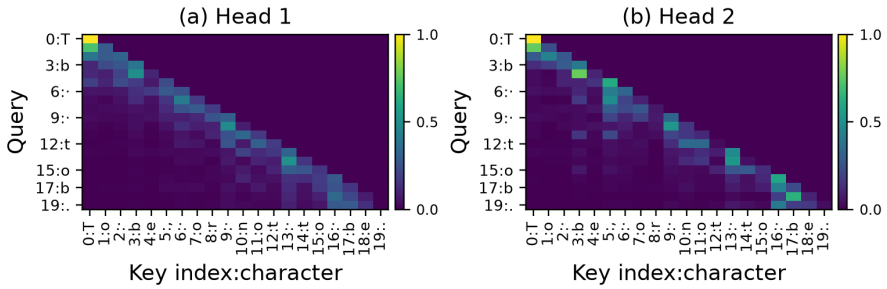


Figure 8. Baseline first-layer heads 1/2 on "To be, or not to be." Rows are queries, columns are keys.

Figure 8 mean diagonal weights are 0.2387 and 0.2671. Future weights are zero; heads distribute weight differently over visible character positions. This single input does not establish semantic specialization or causal importance.

Table 8. GTX 1650 full-model context benchmark

Context	Time(ms), mean ± std	Peak tensor MiB
64	7.1986 ± 0.1017	16.582
128	9.0761 ± 0.0997	37.662
256	16.6287 ± 0.1695	98.960
512	43.8210 ± 0.1007	309.819

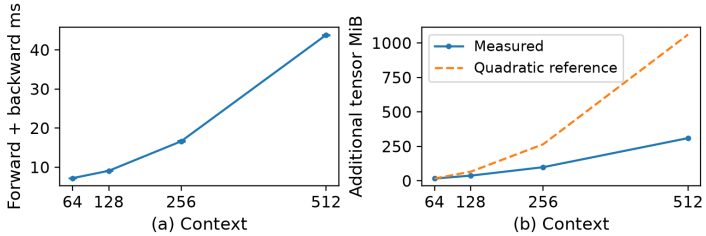


Figure 9. Measured time and incremental allocated tensor memory versus context.

Table 8/Figure 9: batch 8/FP 32, eval-mode forward plus backward, two warmups and five synchronized timings. Peak allocated tensor memory subtracts the model/input baseline. From 64 to 512, time grows 6.09 times and memory 18.68 times. Attention scores grow 64 times; projections and FFNs add linear work, so full-model ratios differ.

Table 9. Bonus: total greedy generation latency on GTX 1650

Prompt+new	Uncached(s)	Cached(s)	Speed ratio
8+64	0.158	0.122	1.30
32+64	0.147	0.121	1.21
64+64	0.142	0.114	1.25
7+500	1.059	1.080	0.98

Table 9 uses selected continued weights, batch 1/FP 32, three warmups and ten alternating paired synchronized trials, including prefill. Ratio = uncached/cached mean latency. Logits pass absolute/relative tolerances 10^{-5}; greedy outputs match exactly. Rebuilding after window shifts preserves original positions and context. Short-prefix speed-ups do not extend to the 500-character rolling case, which is slower. Raw trials are in CSV notes.

References

[1] Course assignment: DNN Programming Assignment–2, RNN, LSTM, GRU and Transformer; current root DOCX identified by the student. Missing variant table and hand fixture are handled as explicit assumptions under its missing-information clause.

[2] Vaswani et al., Attention Is All You Need (2017). https://arxiv.org/abs/1706.03762. The assignment uses pre-LN, unlike the original post-LN block.

[3] UCI Human Activity Recognition Using Smartphones. https://archive.ics.uci.edu/dataset/240/human+activity+recognition+using+smartphones.

[4] Karpathy, Tiny Shakespeare data. https://github.com/karpathy/char-rnn/tree/master/data/tinyshakespeare.

[5] PyTorch reference documentation; recurrent and attention modules used for verification. https://pytorch.org/docs/stable/nn.html.